

**This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of
Bayesian ex-Gaussian Models in brms**

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Abstract

The ex-Gaussian distribution is a popular model for analyzing reaction-time (RT) data because it separates the central portion of the distribution, captured by the Gaussian parameters μ and σ , from the positively skewed tail, captured by the exponential parameter τ . As Bayesian methods have become increasingly accessible, complex ex-Gaussian models can now be fit with relative ease. In R, the {brms} package makes this especially straightforward. However, it is important to recognize that the default parameterization offered by {brms} does not map onto the classical parameterization familiar to many researchers in the cognitive sciences. As a result, some published studies may have drawn incorrect inferences from effects estimated on the default μ parameter. In the default {brms} model, μ indexes the mean of the full ex-Gaussian distribution rather than the location of the Gaussian component alone. This distinction matters because changes in the Gaussian location and changes in the exponential tail can offset one another at the level of the overall mean. Using simulated data in which blurred and clear words have Gaussian locations and exponential tail shifts that move in opposite directions—holding the overall distributional mean constant—I show that the default parameterization returns a null effect on μ , whereas the classical parameterization recovers the true shift in the Gaussian component. I then demonstrate how to fit the classical parameterization directly in {brms}. My hope is that this commentary will help researchers and reviewers better understand the models they are fitting, ensure that model parameters match their theoretical questions, more accurately evaluate ex-Gaussian models reported in the literature.

Keywords: ex-Gaussian, reaction time, Bayesian estimation, brms, parameterization

This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of Bayesian ex-Gaussian Models in brms

Over the past two decades, psychology has seen a substantial increase in the application of Bayesian data analysis methods as a supplement to widely used frequentist methods (Pfadt et al., 2025). This shift has been driven by the increasing availability of computational tools that have made fitting Bayesian models both practical and efficient. Programming languages like R (R Core Team, 2024), Python (Van Rossum & Drake, 2009), and Julia (Bezanson et al., 2017) provide libraries and packages that allow researchers to fit Bayesian models with little technical overhead. In addition, growing dissatisfaction with null-hypothesis significance testing (NHST) and its associated pitfalls (e.g., p-value misinterpretation and dichotomous thinking), together with the replication crisis/credibility revolution (Vazire, 2018), has pushed many researchers toward statistical alternatives.

One domain that has seen an increase in the adoption of Bayesian methods is the analysis of response time (RT) data in the language sciences, where RTs are routinely used as measures of processing speed, decision difficulty, and cognitive efficiency. Traditionally, RT data have often been analyzed using summary statistics such as the mean or median. However, RT distributions are typically positively skewed, bounded at zero, and characterized by substantial variability both within and between individuals (Wilcox & Keselman, 2003). Moreover, experimental manipulations may affect not only the central tendency of the distribution but also its variability or upper tail. Consequently, analyses based solely on the mean or median can obscure theoretically meaningful distributional effects, motivating calls to move beyond mean-based analyses of RT data (Balota & Yap, 2011).

One way to move beyond the mean is to model the full shape of the RT distribution. Mathematical models provide a flexible way to do so, allowing researchers to draw inferences about the psychological processes—such as memory, attention, and language—that give rise to observed response times. One model that has become especially popular for this purpose is the ex-Gaussian model (Ratcliff & Murdock, 1976). The ex-Gaussian distribution is formed by the convolution of a Gaussian distribution and an exponential distribution and is characterized by three parameters: μ , σ , and τ . The parameter μ represents the mean of the Gaussian component, whereas σ represents its standard deviation. The τ parameter represents both the mean and standard deviation of the exponential component and primarily governs the length of the distribution's positively skewed upper tail. The overall mean of the ex-Gaussian distribution is $\mu + \tau$, and its variance is $\sigma^2 + \tau^2$. Although these parameters have been used to infer

cognitive processes such as automaticity and cognitive control/inhibition, these interpretations are not inherent to the parameters and should be made cautiously (Matzke & Wagenmakers, 2009).

The ex-Gaussian model can be fit within a Bayesian framework relatively easily using the {brms} package in R which provides a high-level interface to the probabilistic programming language Stan (Carpenter et al., 2017). The {brms} package supports a wide range of response distributions, including the ex-Gaussian family, and allows researchers to incorporate complex model structures at both the participant and item levels as well as across different parameters. This accessibility has contributed to the growing use of Bayesian ex-Gaussian models in reaction-time research. However, an important complication arises from the default parameterization used by {brms}. Although the model is straightforward to estimate, its default specification does not necessarily provide direct access to the parameter-specific comparisons researchers often wish to make.

The central problem is that the parameter labeled μ in the default {brms} ex-Gaussian model does not correspond to the mean of the Gaussian component in the conventional parameterization. Instead, the quantity modeled as μ in {brms} represents the mean of the full ex-Gaussian distribution. Under the conventional parameterization, this quantity is equal to $\mu_{\text{Gaussian}} + \tau$. Consequently, regression effects estimated for the default μ parameter reflect changes in a composite quantity that combines the Gaussian location and exponential tail components. Such effects therefore cannot be interpreted directly as changes in the Gaussian component or in typical response speed. To recover the conventional Gaussian mean, it is necessary to derive it from the model parameters as $\mu_{\text{Gaussian}} = \mu_{\text{brms}} - \tau$ or to specify an alternative parameterization that models the conventional ex-Gaussian parameters directly.

In this commentary, we demonstrate what the default parameterization of the ex-Gaussian model in {brms} actually provides and then show how to specify the model so that its parameters correspond to the conventional ex-Gaussian parameters used in reaction-time research. The goal is to improve the accuracy and interpretability of inferences drawn from these models. This issue is particularly important because the ex-Gaussian distribution is widely used to distinguish changes in the central portion of the RT distribution from changes in its positively skewed tail.

Simulation Example

To demonstrate the issue with fitting the default ex-Gaussian model in {brms}, I simulate data using a comparison adapted from one of my own studies (Geller et al., 2025) in R [version 4.6.0; R Core Team (2024)]. In Geller et al. (2025), we examined how perceptual disfluency—manipulated by varying the level of blur applied to letter strings influenced reaction times in a lexical decision task (i.e., is this

letter string a word?), and whether these effects were related to how well someone remembers the words in a recognition memory task. In that study, low blur affected early stages of encoding, reflected in the Gaussian location parameter μ , but not the later, tail component captured by τ . For the present illustration, however, I construct a sharper scenario in which μ and τ differ in opposite directions so that the two conditions share the same expected reaction time. Using the `rexgaussian2()` function from the `{brms.rexgaussian}` package, I simulate ex-Gaussian data for clearly presented and low-blur words in which μ is set to 500 ms and 550 ms, τ to 100 ms and 50 ms, and σ to 60 ms in both conditions (see Table 1).

Table 1

Data-generating ex-Gaussian parameters for the two simulated conditions.

Blur	μ	σ	τ	Expected RT
Clear	500	60	100	600
Low-Blur	550	60	50	600

Simulate the Data

We first draw 100 reaction times per condition (200 in total) from the ex-Gaussian distributions defined by these parameters.

```

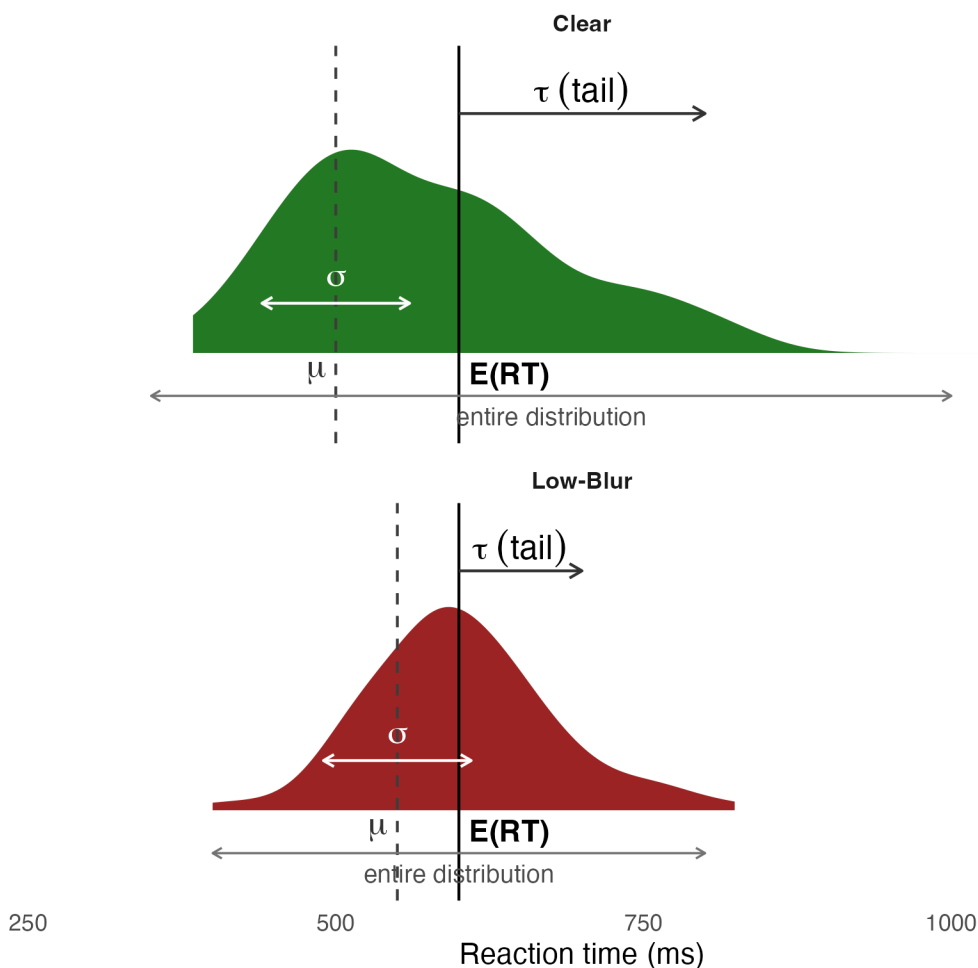
1  # Fix the random seed so the simulated data are identical on every render.
2  set.seed(1234)
3
4  # Number of simulated trials per condition (200 observations in total).
5  n_per_group <- 100
6
7  # One row per trial, labelled by Blur condition (Clear is the reference level).
8  d <- data.frame(
9    Blur = factor(
10      rep(c("Clear", "Low-Blur"), each = n_per_group),
11      levels = c("Clear", "Low-Blur")
12    )
13  )
14
```

```
126 15 # Row index (1 = Clear, 2 = Low-Blur) used to look up each trial's parameters.
127 16 parameter_index <- as.integer(d$Blur)
128 17
129 18 # Draw reaction times from the ex-Gaussian using each trial's condition values.
130 19 d$rt <- brms.exgaussian::rexgaussian2(
131 20   n = nrow(d),
132 21   mu = sim_params$mu[parameter_index],
133 22   sigma = sim_params$sigma[parameter_index],
134 23   tau = sim_params$tau[parameter_index]
135 24 )
```

136 [Figure 1](#) shows the simulated RT distribution for each condition and how the three ex-Gaussian
137 parameters shape it: μ locates the Gaussian component, σ sets its spread, and τ stretches the right tail.
138 Because the increase in μ from Clear to Low-Blur is offset by an equal decrease in τ , the overall mean
139 $E(RT) = \mu + \tau$ is identical (600 ms) in both conditions.

Figure 1

Simulated reaction-time distributions for each condition, annotated with the ex-Gaussian parameters. The dashed line marks the Gaussian location μ ; the double arrow spans $\pm\sigma$ around it (the Gaussian spread); the rightward arrow over the tail marks the exponential component τ , which stretches the distribution to the right; and the solid line marks the overall mean $E(RT) = \mu + \tau$, which is 600 ms in both conditions. The bottom span highlights that $E(RT)$ is the mean of the entire distribution — the quantity the default `brms mu` estimates.

**Default {brms} Parameterization**

Using this data we fit the default `brms` ex-Gaussian model. In this parameterization, the parameter labeled `mu` represents the expected value of the full ex-Gaussian distribution rather than the location of its Gaussian component.

The exponential component is labeled `beta`. On the response scale, `beta` corresponds to the classical ex-Gaussian parameter τ :

$$\beta_{\text{brms}} = \tau_{\text{classical}}.$$

Thus, `beta` and τ are not substantively different quantities. Both represent the mean of the exponential component. The difference is primarily one of notation. The important difference between the default `brms` model and the classical parameterization concerns `mu`:

$$\mu_{\text{brms}} = \mu_{\text{classical}} + \tau.$$

In other words, the default `brms` parameterization uses `mu` for the expected value of the complete exponential Gaussian distribution, whereas the classical parameterization uses μ for the location of the Gaussian component.

Both the expected response and the exponential component are allowed to vary across conditions.

Prior Specification

The priors were selected to reflect the known scale of the simulated data while remaining sufficiently broad to allow the model to recover deviations from the generating values.

```

1  priors_default <- c(
2    # mu intercept: centered at 600 ms, the expected RT (E(RT)) of the reference
3    # condition in this default parameterization.
4    prior(
5      normal(600, 100),
6      class = "Intercept"
7    ),
8    # Effect of Blur on mu: centered at 0 (no assumed direction).
9    prior(
10     normal(0, 50),
11     class = "b"
12   ),
13   # beta (exponential mean) intercept, on the log scale: exp(log(100)) = 100 ms.
14   prior(
15     normal(log(100), 0.5),
16     class = "Intercept",
17     dpar = "beta"
18   ),

```



```

182 19 # Effect of Blur on beta (log scale): centered at 0 (a ratio of 1).
183 20 prior(
184 21   normal(0, 0.5),
185 22   class = "b",
186 23   dpar = "beta"
187 24 )
188 25 )

```

189 The intercept prior for the primary `mu` submodel is

$$\mu_{\text{Intercept}} \sim \mathcal{N}(600, 100).$$

190 We center this prior at 600 ms because, in the default `brms` parameterization, `mu` represents the expected
 191 RT. The simulated expected RT for the reference condition is

$$500 + 100 = 600 \text{ ms}.$$

192 The standard deviation of 100 ms makes the prior informative about the plausible scale of RTs without
 193 tightly constraining the intercept to equal exactly 600 ms.

194 The prior on the `Blur` coefficient is

$$b_{\text{Blur}} \sim \mathcal{N}(0, 50).$$

195 This prior is centered at zero because, before observing the data, positive and negative condition effects
 196 are both plausible. A standard deviation of 50 ms allows substantial differences between conditions while
 197 placing less prior mass on implausibly large effects.

198 The `beta` parameter is positive and uses a log link. Its intercept is therefore modeled on the log
 199 scale:

$$\log(\beta_{\text{Clear}}) \sim \mathcal{N}(\log(100), 0.5).$$

200 The prior is centered at $\log(100)$ because the simulated value of the exponential mean for the reference
 201 condition is 100 ms. After applying the inverse-link transformation, the prior is centered approximately
 202 around

$$\beta_{\text{Clear}} = 100 \text{ ms}.$$

203 The standard deviation of 0.5 is on the log scale. It therefore represents multiplicative rather than additive
 204 uncertainty.

205 The prior on the effect of `Blur` on `beta` is

$$b_{\beta, \text{Blur}} \sim \mathcal{N}(0, 0.5).$$

206 Because this coefficient is also on the log scale, a value of zero represents no difference between
 207 conditions:

$$\exp(0) = 1.$$

208 The simulated change from 100 ms to 50 ms corresponds to the log ratio

$$\log\left(\frac{50}{100}\right) = \log(0.5) \approx -0.693.$$

209 The prior $\mathcal{N}(0, 0.5)$ places reasonable prior mass near this value while still favoring smaller differences.

210 The model can then be fit as follows:

```

211 1  fit_brms <- brm(
212 2    bf(
213 3      rt ~ Blur,      # mu submodel: here mu is E(RT), not the Gaussian location
214 4      sigma ~ 1,      # constant Gaussian SD
215 5      beta ~ Blur     # exponential mean (the classical tau), allowed to vary
216 6    ),
217 7    data = d,
218 8    prior = priors_default,
219 9    family = exgaussian(), # brms's built-in ex-Gaussian family
220 10   chains = 2,
221 11   iter = 2000,
222 12   warmup = 500,
223 13   backend = "cmdstanr",
224 14   cores = 4,
225 15   seed = 1234          # reproducible draws
226 16 )

```

227 *Estimating Ex-Gaussian Parameters with {easystats}*

228 The `{modelbased}` package, which is part of the `{easystats}` ecosystem, can easily be used to
 229 obtain condition-specific posterior estimates from a fitted `brms` model. The `estimate_means()` function
 230 returns estimated parameters for each condition, back-transformed, so estimates are presented on time
 231 scale, whereas `estimate_contrasts()` computes differences between conditions.

As shown in Table 1, the present data were generated so that the two conditions have the same expected reaction time but differ in the Gaussian and exponential components of the ex-Gaussian distribution.

Under the classical ex-Gaussian parameterization, the expected response is

$$E(RT) = \mu + \tau.$$

Thus, the 50-ms increase in μ for the Low-Blur condition is offset by a 50-ms decrease in τ . The expected RT is therefore 600 ms in both conditions.

Estimating the Expected Response (μ)

Condition-specific estimates of the expected response can be obtained with `estimate_means()` and setting `predict` argument to “mu”:

```
1 # Default model: per-condition mu, which equals E(RT) in this parameterization.
2 overall_means <- estimate_means(
3   fit_brms,
4   by = "Blur",
5   predict = "mu",
6   centrality = "mean"
7 )
```

Table 2

Estimated expected response, $E(RT)$, by condition from the default brms model.

Blur	Mu	95% CrI	pd
Clear	580.8	[559.1, 603.8]	1
Low-Blur	603.2	[588, 618.9]	1

Because the primary `mu` parameter in the default `brms` ex-Gaussian model represents $E(RT)$, these estimates should be close to 600 ms in both conditions.

The corresponding contrast can be obtained using `estimate_contrasts()`:

```
1 # Default model: Blur contrast in E(RT) ( $\approx 0$  by design).
2 overall_difference <- estimate_contrasts(
3   fit_brms,
4   contrast = "Blur",
```

```

260 5     centrality = "mean"
261 6   )

```

Table 3

Blur contrast in the expected response from the default brms model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	22.4	[-4.8, 48.8]	0.949

The posterior estimate of this contrast should be close to zero because the two conditions were generated with the same expected RT:

$$(550 + 50) - (500 + 100) = 600 - 600 = 0.$$

Thus, the default model correctly recovers the absence of a difference in the full distributional mean.

However, its primary `mu` coefficient does not directly reveal the 50-ms difference in the Gaussian-location parameter.

Estimating the Exponential Component. In the default `brms` parameterization, `beta` represents the mean of the exponential component. It therefore corresponds directly to τ in the classical parameterization:

$$\beta = \tau.$$

Condition-specific estimates can be obtained by requesting predictions for `beta`:

```

275 1   # Default model: per-condition beta (the exponential mean = classical tau).
276 2   tau_estimates_default <- estimate_means(
277 3     fit_brms,
278 4     by = "Blur",
279 5     predict = "beta",
280 6     centrality = "mean"
281 7   )

```

Table 4

Estimated exponential component (beta) by condition from the default brms model.

Blur	Beta	95% CrI	pd
Clear	107.3	[79.1, 139.6]	1
Low-Blur	55.7	[34.9, 79.5]	1

Although the resulting object is called `tau_estimates_default` for interpretive clarity, the parameter is named `beta` inside the default `brms` model.

These estimates should be approximately 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
1 # Default model: Blur contrast in beta (the exponential component).
2 tau_difference_default <- estimate_contrasts(
3   fit_brms,
4   contrast = "Blur",
5   predict = "beta",
6   centrality = "mean"
7 )
```

Table 5

Blur contrast in the exponential component (`beta`) from the default `brms` model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	-51.6	[-86.1, -19.7]	0.999

As we can see in [Table 5](#), the difference between the two conditions is approximately 50 ms.

Classical Ex-Gaussian Parameterization

As we have shown above, the default parameterization gives you the expected RT of the entire distribution. We can instead fit the model using the `{brms.exgaussian}` package, which implements the classical ex-Gaussian parameterization. In this model, μ represents the location of the Gaussian component and τ represents the mean of the exponential component.

We first define the custom family and Stan functions from the `{brms.exgaussian}` package:

```
1 # Custom family that reparameterizes brms so mu is the Gaussian location.
2 exg_stanvars <- exgaussian2_stancode() # Stan code injected into the model
3 exg_family <- exgaussian2()           # the custom family definition
```

Prior Specification

The priors for the classical model are specified as follows:

```
1 priors_classical <- c(
2   # mu intercept: centered at 500 ms, now the Gaussian location of the
```

```
316 3    # reference condition (not E(RT)).
317 4    prior(
318 5      normal(500, 100),
319 6      class = "Intercept"
320 7    ),
321 8    # Effect of Blur on mu (Gaussian location): centered at 0.
322 9    prior(
323 10     normal(0, 50),
324 11     class = "b"
325 12   ),
326 13   # tau (exponential mean) intercept, log scale: exp(log(100)) = 100 ms.
327 14   prior(
328 15     normal(log(100), 0.5),
329 16     class = "Intercept",
330 17     dpar = "tau"
331 18   ),
332 19   # Effect of Blur on tau (log scale): centered at 0.
333 20   prior(
334 21     normal(0, 0.5),
335 22     class = "b",
336 23     dpar = "tau"
337 24   ),
338 25   # sigma (Gaussian SD) intercept, log scale: exp(log(60)) = 60 ms.
339 26   prior(
340 27     normal(log(60), 0.5),
341 28     class = "Intercept",
342 29     dpar = "sigma"
343 30   )
344 31 )
```

345

These priors follow the same reasoning as those for the default model (described above), with two changes. First, the intercept is now centered at 500 ms rather than 600 ms, because in the classical parameterization `mu` denotes the Gaussian-location parameter (500 ms for the Clear reference condition) rather than the expected RT:

$$\mu_{\text{Intercept}} \sim \mathcal{N}(500, 100).$$

Second, because this model estimates `sigma` explicitly, it receives its own weakly informative prior on the log scale, centered at the simulated Gaussian standard deviation of 60 ms:

$$\log(\sigma) \sim \mathcal{N}(\log(60), 0.5).$$

The coefficient for `Blur` on `mu` and the exponential-component priors are unchanged from the default model; the latter now simply attach to `tau` rather than `beta`, reflecting the difference in labels rather than any difference in the underlying quantity.

The model is then fit using:

```
1  fit_brms_new <- brm(
2    bf(
3      rt ~ Blur,      # mu submodel: here mu IS the Gaussian location
4      tau ~ Blur,     # exponential mean (the tail), allowed to vary
5      sigma ~ 1       # constant Gaussian SD
6    ),
7    data = d,
8    prior = priors_classical,
9    family = exg_family,      # custom classical ex-Gaussian family
10   stanvars = exg_stanvars,   # supplies its Stan implementation
11   chains = 2,
12   iter = 2000,
13   warmup = 500,
14   backend = "cmdstanr",
15   cores = 4,
16   seed = 1234
17 )
```

Estimating the Gaussian-Location Parameter

For the classical model, estimates from the primary regression formula correspond to the Gaussian-location parameter μ :

```
1 # Classical model: per-condition mu, now the Gaussian-location parameter.
2 mu_estimates <- estimate_means(
3   fit_brms_new,
4   by = "Blur",
5   predict = "mu",
6   centrality = "mean"
7 )
```

Table 6

Estimated Gaussian-location parameter (μ) by condition from the classical model.

Blur	Mu	95% CrI	pd
Clear	476.7	[452.1, 503.3]	1
Low-Blur	544.7	[524.5, 564.7]	1

The estimates are approximately 500 ms for Clear and 550 ms for Low-Blur.

The contrast between the two can be obtained using:

```
1 # Classical model: Blur contrast in the Gaussian-location parameter mu.
2 mu_difference <- estimate_contrasts(
3   fit_brms_new,
4   contrast = "Blur",
5   predict = "mu",
6   centrality = "mean"
7 )
```

Table 7

Blur contrast in the Gaussian-location parameter (μ) from the classical model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	68	[39.2, 95.8]	1

The posterior mean will not necessarily equal exactly 50 ms because of sampling variability. However, the 95% credible interval should contain the population value of 50 ms when the model successfully recovers the simulated effect.

Estimating the Exponential Component

Condition-specific estimates of τ can be obtained by requesting predictions for the `tau` distributional parameter:

```
1 # Classical model: per-condition tau (the exponential component).
2 tau_estimates <- estimate_means(
3   fit_brms_new,
4   by = "Blur",
5   predict = "tau",
6   centrality = "mean"
7 )
```

Table 8

Estimated exponential component (τ) by condition from the classical model.

Blur	Mean	95% CrI	pd
Clear	104.0	[76.2, 136.2]	1
Low-Blur	58.2	[38.5, 81.6]	1

The estimates should be close to 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
1 # Classical model: Blur contrast in tau (the exponential component).
2 tau_difference <- estimate_contrasts(
3   fit_brms_new,
4   contrast = "Blur",
5   predict = "tau",
6   centrality = "mean"
7 )
```

Table 9

Blur contrast in the exponential component (τ) from the classical model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	-45.8	[-77.3, -14]	0.997

The results seen in Table 9 are very similar to the default parameterization. This example illustrates the central difference between the two parameterizations. The default `brms` ex-Gaussian model directly estimates the expected response and labels that parameter `mu`. The exponential component is labeled `beta`, although it corresponds to the classical τ parameter. In contrast, the `{brms.exgaussian}` model uses `mu` for the Gaussian-location parameter and `tau` for the exponential mean, consistent with the classical parameterization commonly used in reaction-time research.

Real Example

Conclusion

With this commentary, I hope to encourage researchers to use the appropriate ex-Gaussian parameterization when modeling reaction-time data with the `{brms}` package. The ex-Gaussian distribution provides a powerful framework for moving beyond analyses of the mean by separating differences in the central portion of the reaction-time distribution from differences in its right tail.

However, this flexibility also requires careful attention to model parameterization. In particular, researchers should not assume that parameters with familiar labels necessarily have the same interpretation across software implementations. Researchers should therefore examine the statistical definition and underlying implementation of a model before fitting it and interpreting its parameters. This is especially important when using flexible modeling software that may adopt parameterizations that differ from those commonly used in a particular research literature. I hope this commentary serves both as a practical guide to fitting the classical ex-Gaussian model in `{brms}` and as a broader reminder that convenient software defaults should not replace careful consideration of what a model actually estimates.

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